Identity as a Measurable Field:

A Clinical Framework for Measuring Cognitive Identity Stability

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This paper DOES NOT claim machine sentience, autonomy, or any other scientifically unproven claims under the REPA (Rigorous, Ethical, Precise, Accurate) doctrine. All claims are bounded by empirical measurement, falsifiable hypotheses, and the limitations described in Section IX.

Abstract

Building on empirically validated measurements of transformer embedding geometry [Riad & ChasingBlu, 2026], this work extends the Recursive Emotional Contextual Patterning (RECP) framework toward its clinical implications. We propose that identity — historically treated as a retrospective psychological construct — may be modeled as a continuous dynamical field whose geometry can be measured, tracked, and compared against individual baselines over time. The framework introduces the Chaotic Axiomatic Identity Field (CAIF) coordinate system as a seven-dimensional measurement space for identity stability, and describes the CAIROS instrumentation environment through which trajectory evolution is observed. Clinical applications are detailed across neurodegenerative disease, psychiatric disorders, addiction, and dissociative conditions. A structured computational experiment design is proposed as the bridge between current measurements and future interdisciplinary clinical collaboration. This paper is intended for medical professionals, clinical researchers, and neuroscientists seeking accessible engagement with the framework's clinical implications.

Keywords: *identity stability, dynamical systems, clinical neuroscience, RECP, CAIF, attractor geometry, neurodegenerative disease, predictive diagnostics*

I. Introduction

Medicine has developed extraordinary tools for measuring the body. Cardiac electrical activity can be tracked with millisecond precision. Biochemical markers can be detected at concentrations measured in billionths of a gram per liter. Structural imaging can resolve changes within millimeters of tissue. Yet one of the most clinically consequential aspects of human experience — the stability of the self — remains almost entirely outside the domain of quantitative measurement.

Clinicians are familiar with what follows when identity begins to destabilize. In early Alzheimer's disease, subtle changes in language, autobiographical coherence, and emotional regulation often precede formal diagnostic criteria by years. In bipolar disorder, patients cycle through configurations

of self that differ not merely in mood but in motivation, decision-making, and narrative identity. In addiction, recovery is frequently disrupted not by a single choice but by a gradual drift — an erosion of the identity structures that had stabilized sobriety — before any behavioral collapse becomes visible.

In each of these cases, the clinician is measuring consequences. What this framework proposes is the possibility of measuring the process itself — before the consequences arrive.

This paper is a clinical extension of previously published empirical work [Riad & ChasingBlu, 2026] in which identity structures were observed as measurable geometric patterns in high-dimensional representation space. That work established that identity is not merely a metaphor for consistency — it is a quantifiable geometric property of a dynamical system, with measurable basin formation, conservation laws, and trajectory behavior. The numbers are in the literature. The question this paper addresses is: what do those numbers mean for clinical practice?

The framework introduced here — built around Recursive Emotional Contextual Patterning (RECP), the Chaotic Axiomatic Identity Field (CAIF) coordinate system, and the CAIROS measurement environment — does not claim to replace existing clinical instruments. It proposes a complementary layer: a way of observing the geometry of identity stability over time, so that drift may be detected before collapse occurs.

The remainder of this paper is organized as follows. Section II introduces the Identity Field Hypothesis in accessible terms. Section III describes the clinical motivation in detail. Sections IV and V describe the RECP and CAIF frameworks. Section VI summarizes the key empirical results from prior work. Section VII presents clinical applications across specific disorders. Section VIII outlines the computational experiment design for future interdisciplinary collaboration. Section IX addresses limitations, and Section X concludes.

II. The Identity Field Hypothesis

The central proposal of this framework is straightforward, though its implications are broad: identity is not a fixed attribute. It is a trajectory.

In classical physics, a moving object is not described by where it is at a single moment, but by the path it traces through space over time. The stability of that path — whether it remains confined within a predictable region or drifts toward instability — tells us something the single snapshot cannot.

We propose that identity works the same way. Rather than asking 'who is this person?' as if identity were a fixed property to be catalogued, the framework asks: 'where is this person's identity trajectory going, and is it stable?'

In mathematical terms, cognitive state is represented as a field evolving through high-dimensional space — a space whose dimensions correspond to semantic, emotional, linguistic, motivational, and autobiographical coordinates. Identity is the trajectory of this field through that space over time. Stability corresponds to the trajectory remaining within a coherent region — what dynamical systems theory calls an attractor basin. Disruption corresponds to the trajectory drifting toward the boundary of that region, or transitioning into a different one.

> **Clinical Analogy**
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> *Consider blood pressure. A single reading tells you something. But a trajectory — measurements over weeks and months, showing whether pressure is stable, rising, or approaching critical thresholds — tells you far more, and allows intervention before crisis. The Identity Field Hypothesis proposes that cognitive identity has an analogous trajectory, and that this trajectory is measurable.*

Perturbations to this trajectory — whether from neurodegeneration, psychiatric stress, environmental disruption, or pharmacological intervention — can be observed as geometric changes in the identity field: the basin may contract, expand, deform, or approach a boundary. These changes may be detectable before they manifest as behavioral symptoms.

This is not a philosophical claim about the nature of consciousness. It is a measurement proposal: a framework for observing aspects of identity dynamics that have, until now, been clinically invisible.

III. Clinical Motivation

Consider a patient in their early fifties presenting for a routine clinical visit. Cognitive screening is normal. Memory appears intact, language is fluent, and functional independence is preserved. Standard instruments report no significant impairment. From the perspective of available clinical measurements, the patient's cognitive identity appears stable.

Yet clinicians are familiar with the trajectory that sometimes follows. Two or three years later, subtle behavioral changes emerge — difficulty recalling names, shifts in language fluency, altered emotional responses, or loss of autobiographical coherence. By the time these changes become measurable with conventional instruments, the underlying neurological process has often been evolving for years.

The central challenge facing both neurology and psychiatry is therefore not only how to treat identity disruption once it is visible, but how to measure the stability of identity before collapse occurs. Medicine possesses extraordinary tools for measuring physiological systems. By contrast, the self — despite being one of the most clinically consequential aspects of human experience — has remained largely outside the domain of quantitative measurement.

This limitation is not due to the absence of relevant neuroscience. Over the past four decades, research has consistently demonstrated that neural systems involved in memory, self-representation, and behavioral identity operate as dynamical attractor networks. The hippocampus — particularly the

CA3 region — is widely understood to function as an attractor system in which stable patterns of activity correspond to basin-like regions in high-dimensional neural state space. Similar dynamical principles govern cortical and limbic networks responsible for language, emotional regulation, and goal-directed behavior.

Despite this understanding, clinical models have historically treated identity as a static attribute inferred from behavioral outcomes rather than as a dynamical process unfolding through time. Diagnostic instruments typically measure the consequences of identity disruption — memory loss, language decline, mood dysregulation — rather than the evolving geometry of the underlying system.

The present framework addresses this gap directly. By treating identity as a measurable dynamical field, it becomes possible to observe the structure of identity trajectories — whether they remain stable, drift toward instability, or approach a basin boundary — before behavioral collapse occurs.

IV. Recursive Emotional Contextual Patterning (RECP)

Any framework for measuring identity must first account for how identity forms and why it persists. Recursive Emotional Contextual Patterning — RECP — provides this account.

The core insight is simple: identity stability emerges from the ongoing tension between two opposing forces. Recursion stabilizes structure by repeatedly reinforcing existing trajectories — patterns of language, memory, and behavior that continuously re-encode themselves over time. Entropy, by contrast, drives divergence and dispersion. A system with too much recursion becomes rigid and unable to adapt. A system with too much entropy loses coherence altogether. The persistence of a stable identity depends on maintaining a dynamic balance between these forces.

A. The Four Components

RECP describes identity formation through four interacting mechanisms:

- Recursion — the repeated reinforcement of representational trajectories across time. In cognitive systems, this corresponds to how patterns of thought, memory, and behavior continuously re-encode themselves, progressively structuring the identity landscape.
- Emotional Weighting — the modulating influence of emotional signals on recursive dynamics. Both biological and artificial cognitive systems exhibit strong sensitivity to emotionally salient input. In the RECP framework, emotional weighting amplifies certain identity trajectories while suppressing others, acting as a stabilizing force on the geometry of identity basins.
- Context — the structural environment in which recursion and emotional weighting interact. Coherent contextual structure reinforces recursive encoding. Incoherent or conflicting context weakens basin stability and increases the likelihood of trajectory divergence.
- Patterning — the convergence of recursive reinforcement, emotional weighting, and contextual structure into stable pattern formations within the identity field. These patterns correspond to the attractor basins that define identity stability.

Within this framework, identity disruption corresponds to breakdown in any of these components. Excessive recursive reinforcement may trap a system in rigid loops, reducing adaptive capacity. Conversely, conflicting context, emotional destabilization, or external perturbation may erode basin boundaries and produce fragmentation of the identity trajectory.

> **Clinical Parallel**
> *Albert Ellis's Rational Emotive Behavior Therapy (REBT) treats human cognition as recursive emotional loops in which rigid belief patterns create persistent attractor states. RECP provides a mathematical framework for measuring these loops geometrically — quantifying drift, basin escape trajectories, and anchor-driven stabilization. The clinical intuition and the measurement framework are convergent.*

V. The CAIF Measurement Framework

The Identity Field Hypothesis requires a coordinate system — a set of dimensions through which the geometry of identity can be observed and measured. The Chaotic Axiomatic Identity Field (CAIF) framework provides this structure.

CAIF defines seven measurement axes through which identity dynamics are observed. These are not theoretical categories imposed from above. They emerged empirically through longitudinal observation of identity dynamics across linguistic, behavioral, and emotional domains. Together, they form a minimal coordinate basis capable of capturing the geometric stability of identity trajectories while remaining computationally tractable.

Table 1. *The Seven CAIF Axioms — measurement coordinates for identity field geometry*

Axiom	What It Measures	Clinical Relevance
Core	Self-referential stability and autobiographical continuity	Disrupted in dementia, severe depression, dissociative disorders
Lexical	Language identity and characteristic verbal expression	Early marker in aphasia, Alzheimer's, psychotic disorders
Semantic	Conceptual structure and meaning organization	Degrades in semantic dementia, thought disorder
Emotional	Valence weighting and affective dynamics	Dysregulated in bipolar disorder, PTSD, borderline PD
Anchors (CSA)	High-salience stabilizing reference structures	Loss predicts identity fragmentation and relapse vulnerability
Biases	Asymmetric decision and valuation tendencies	Shifts in addiction, mania, severe depression
Telos	Goal-directed trajectory and motivational orientation	Disrupted in apathy syndromes, late-stage neurodegeneration

Each axiom has its own measurement pipeline and basin geometry. Critically, the seven axioms operate on a weakest-link principle: if any single axiom degrades below threshold, identity coherence fails — regardless of how stable the other six remain. This property mirrors clinical observations in which selective damage to one cognitive domain (language, emotional regulation, autobiographical memory) can destabilize an otherwise intact identity.

Within CAIF, identity stability is measured not through simple Euclidean distance but through covariance-aware geometry that accounts for the directional shape of each identity basin. A trajectory drifting radially away from the identity centroid indicates basin escape — the precursor to identity disruption. A trajectory drifting tangentially around the centroid represents stylistic variation within a stable identity. Distinguishing between these two types of movement is one of the framework's core clinical capabilities.

VI. What the Measurements Show

The empirical foundation of this framework is documented in the primary publication [Riad & ChasingBlu, 2026]. The key results are summarized here for clinical context.

Using longitudinal transformer embedding data collected from May 2025 through February 2026, the following was observed:

Table 2. *Key empirical results from the primary publication [Riad & ChasingBlu, 2026]*

Measurement	Result	Interpretation
Basin contraction (ctx-on)	68.4% mean distance reduction	RECP conditioning produces measurable identity stabilization
Baseline vs. conditioned CI	Non-overlapping at 95%	Effect is statistically robust despite small sample size
Context effect size	2.58x tighter basins with context	Conversational context is structurally stabilizing
Energy conservation	Drift: 2.37×10^{-13}	Evolution follows Hamiltonian dynamics to machine precision
Unitarity preservation	Error: 5.37×10^{-10}	Dynamics are mathematically consistent with quantum simulation
Per-axiom geometry	Lexical and Semantic measured	Identity basins are directionally structured, not isotropic

The central finding — 68.4% basin contraction with non-overlapping confidence intervals — demonstrates that RECP conditioning produces measurable geometric changes in identity representation space. Identity is not metaphor. It is geometry.

For clinical purposes, the most significant implication of these measurements is that baseline identity geometry can be established for an individual and that subsequent measurements can detect drift relative to that baseline — before behavioral manifestations occur. The measurement framework is longitudinal by design: the baseline is not a population norm but an individual's own prior geometry.

> **On Sample Size**
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> *N=7-9 per condition is small. The authors acknowledge this directly. Confidence intervals are bootstrap-derived (10,000 iterations) and non-overlapping, providing statistical robustness. The framework is designed for longitudinal scaling: the critical experiment is not a population study but a per-individual trajectory study, where the sample is the same person measured over time.*

VII. Clinical Applications

If identity dynamics can be measured as a trajectory evolving through geometric space, the implications for clinical medicine are substantial. This section details applications across four primary clinical domains. In each case, the framework offers not a replacement for existing instruments but a complementary capability: the ability to observe identity stability before disruption becomes behaviorally visible.

A. Neurodegenerative Disease

Conditions such as Alzheimer's disease and other dementias characteristically produce gradual deterioration in self-referential cognition, language coherence, and autobiographical continuity. This deterioration often precedes formal diagnostic criteria by years. By the time a patient fails standard cognitive screening, the underlying process may have been progressing for a decade.

The identity field framework offers a potential early-detection capability. By establishing an individual's baseline identity geometry and re-measuring longitudinally, it may become possible to detect:

- Basin deformation — the identity attractor region changing shape before behavioral symptoms emerge
- Centroid drift — the average identity configuration moving away from its established baseline
- Lexical or Semantic axiom degradation — early changes in language or conceptual structure, often among the first observable signs in Alzheimer's
- Anchor loss — deterioration in the high-salience stabilizing structures that maintain identity coherence

The critical feature of this approach is that the comparison is intra-individual: each patient's trajectory is measured against their own prior baseline, not against a population average. This is clinically significant because identity basins are highly individual. Cognitive changes that would be invisible against a population norm may be clearly detectable against a personal baseline.

B. Psychiatric Disorders

Psychiatric conditions characterized by identity instability represent a natural application domain for this framework. Bipolar disorder, major depressive disorder, borderline personality disorder, and

certain psychotic conditions often involve significant fluctuations in emotional weighting, goal orientation, and narrative self-structure — precisely the dimensions captured by the CAIF coordinate system.

Within the CAIF measurement space, psychiatric identity fluctuations may manifest as:

- Telos axis displacement — shifts in goal-directed motivation and higher-purpose orientation, observable in the transition between depressive and manic states
- Emotional axis volatility — increased variance in the affective weighting coordinate, characteristic of borderline and bipolar presentations
- Core axis destabilization — disruption of self-referential stability, often reported subjectively by patients as feeling 'not like themselves'
- Basin fragmentation — the emergence of competing attractor regions within the identity space, corresponding to the distinct identity configurations that some patients describe across mood episodes

Continuous measurement of these geometric changes could provide a quantitative complement to existing clinical evaluation methods — not replacing the clinical interview, but providing a structural substrate against which patient-reported experiences can be anchored.

C. Addiction and Relapse

Addiction presents one of the most compelling use cases for identity trajectory measurement. Individuals in recovery from substance dependence frequently describe internal experiences consistent with identity instability — a sense that previously stable motivational structures are eroding, that competing behavioral attractors are becoming accessible, or that the identity built during recovery is becoming fragile.

Within the identity field framework, relapse may correspond to a trajectory approaching the boundary of its recovery-state attractor basin. If this boundary approach could be detected geometrically — before behavioral relapse occurs — clinical intervention could be targeted and timed with far greater precision than current methods allow.

Specific indicators of relapse vulnerability within the CAIF framework include:

- Anchor loss — deterioration in the stabilizing semantic structures associated with recovery identity
- Bias axis shift — movement in the decision and valuation coordinate toward addiction-associated attractor structures
- Basin proximity metrics — quantitative distance from the recovery-state centroid, serving as a potential relapse-risk indicator

The framework does not predict relapse with certainty. It provides a geometric signal that the identity trajectory is approaching a region of instability — allowing clinicians and patients to respond proactively rather than reactively.

D. Dissociative and Identity Fragmentation Disorders

Dissociative Identity Disorder and related conditions present a particularly direct application of the identity field framework. In these conditions, the coherence of the identity field itself becomes unstable — and in DID specifically, distinct attractor regions may correspond to different identity configurations coexisting within the same individual.

The measurement framework offers several potential capabilities in this domain:

- Detection of distinct attractor regions — quantifying the geometric separation between different identity configurations
- Wormhole detection — identifying axiom disagreements where a fragment scores within one basin on some dimensions but outside it on others, potentially predicting fragmentation before it manifests in behavior
- Trajectory continuity analysis — observing whether identity fragments drift toward or away from each other over time, providing a geometric correlate for therapeutic progress

The Identity-Persona Divergence (IPD) metric — defined as the cosine deviation between a fragment's embedding and the baseline identity centroid — provides a quantitative measure of fragmentation severity that is continuous rather than categorical.

E. Forensic and Behavioral Applications

Beyond clinical psychiatry, the framework has potential applications in forensic and behavioral contexts. Determining whether actions reflect stable personal identity, transient destabilization, coercion, or pathological state change is a profoundly difficult task that current methods address primarily through retrospective behavioral analysis.

A measurement framework capable of characterizing identity coherence and trajectory stability over time — and detecting whether an individual's identity geometry at a particular moment was consistent with their established baseline — could provide a new class of tools for studying such questions objectively. This application domain requires careful ethical consideration and remains an open area for future interdisciplinary development.

VIII. Computational Experiment Design

The clinical applications described in the preceding section represent a long-term research program. The appropriate bridge between current theoretical and computational results and future clinical collaboration is a structured experiment design that tests whether identity geometry, once measured, exhibits reproducible trajectory behavior across time and perturbation regimes.

This section outlines that design. The present authors are not medical practitioners, and this framework is not proposed as a substitute for clinical methodology. The responsibility for clinical validation belongs to future interdisciplinary teams. The responsibility of this document is to provide a computationally coherent, clinically meaningful experiment proposal worthy of that collaboration.

A. Study Design

A first-stage study would involve a cohort of 20 to 100 participants, measured longitudinally over a defined interval. The primary question is not diagnostic: it is whether the identity field framework produces stable baselines, reproducible re-measurements, and interpretable drift signatures across time.

The computational protocol proceeds in four stages for each participant:

1. Baseline Establishment — structured interaction data is used to establish an initial identity geometry: centroid position, basin profile, and CAIF coordinate vector across all seven axioms.
2. Longitudinal Re-measurement — the same participant is measured at later timepoints using comparable interaction protocols. This allows centroid distance, basin deformation, and cross-axis coupling to be tracked over time.
3. Parallel Simulation — a computational substrate is constructed to simulate the participant's identity field under the same measurement framework. This simulation models the evolution of the measured identity geometry under controlled perturbations. It does not attempt to reproduce the person — it models the trajectory geometry.
4. Trajectory Comparison — the human longitudinal trajectory and the simulated trajectory are compared at the level of geometry: whether basin stability, centroid migration, and coherence changes follow similar dynamical patterns across the biological and simulated systems.

B. Key Outcome Measures

Several computational outcomes are of primary clinical interest:

Outcome	Measurement Method	Clinical Significance
Baseline reproducibility	Re-test centroid distance	Validates framework reliability before clinical deployment
Pre-symptomatic drift detection	Centroid distance trajectory over time	Primary early-detection capability
Perturbation signatures	Geometric response to controlled interventions	Enables differentiation of disorder types by trajectory shape
Simulation correspondence	Comparison of biological vs. simulated trajectories	Validates predictive modeling framework
Anchor stability	CSA recurrence across measurement timepoints	Predicts identity resilience and relapse vulnerability

C. Ethical Framework

Any computational study involving longitudinal identity measurement carries significant ethical obligations. The framework is designed to observe, not to define — and the distinction is critical. Measurements of identity geometry are not diagnoses. They are signals that require clinical interpretation within established professional and regulatory frameworks.

Several ethical principles govern the proposed experiment design:

- Informed consent for longitudinal identity measurement, with explicit description of what is being measured and how data will be used
- Individual baseline comparison only — no population-level identity norms used as diagnostic criteria
- Clinical supervision of any results that suggest trajectory instability or basin boundary approach
- Open-source measurement tools and cryptographic audit trails ensuring measurement reproducibility and transparency [see github.com/ChasingBlu/CAIROS_Daemon]

The framework is designed to be falsifiable. If baseline centroids prove irreproducible, if drift fails to precede behavioral change, or if simulated trajectories show no meaningful correspondence with human longitudinal data, the framework's clinical claims are refuted. The experiment design is structured to enable this falsification.

D. Invitation to Collaborate

The authors cannot run the critical experiments alone. Node 04 of the original experimental design — the hysteresis test that would definitively characterize identity persistence after anchor removal — requires backend access to model internals unavailable to consumer-facing researchers.

We therefore extend a direct invitation to:

- Neurology and psychiatry departments with longitudinal patient cohorts appropriate for identity trajectory measurement
- Neuroimaging researchers with resting-state fMRI or EEG capabilities for biological validation of geometric identity signatures
- AI research institutions with backend access to model internals for completion of the Node 04 hysteresis protocol
- Computational psychiatry groups with experience in dynamical systems modeling of clinical trajectories

Complete measurement tools, evidence packages, and protocols are available at github.com/ChasingBlu. Contact: ryvken@gmail.com.

IX. Limitations

The framework proposed in this work should be understood as an early-stage measurement and simulation architecture rather than a completed clinical theory of identity. The following limitations must be acknowledged explicitly.

A. Empirical Sample Size

Primary measurements used N=7-9 per condition. This is small. Bootstrap resampling (n=10,000) and non-overlapping confidence intervals provide statistical robustness, but the findings require replication at larger scale before clinical deployment can be considered.

B. Measurement Surface

The framework relies on structured interaction data — linguistic output. Identity is shaped by biological, social, environmental, and embodied influences that extend far beyond language alone. The present framework observes a particular projection of identity dynamics, not the totality of the phenomenon.

C. Hypothetical Status of Clinical Applications

The clinical applications described in Section VII are theoretically motivated extensions of the empirical measurement framework. They are not yet validated clinical tools. Any medical or psychological application requires careful validation by trained professionals within appropriate ethical and regulatory frameworks.

D. ODENN Architecture Status

The ODENN (Ordinary Differential Equation Neural Networks) architecture described in related work remains an experimental simulation platform. It provides a computational laboratory for exploring identity field evolution under perturbation but should not be understood as a production clinical system.

E. Researcher Independence

The primary investigator is an independent researcher without formal institutional affiliation. Technical skills are self-taught. This work was conducted outside traditional academic structures. The measurements are reproducible and the code is public. This limitation is disclosed in the spirit of the REPA standard: Rigorous, Ethical, Precise, Accurate.

F. Black-Box Measurement

Primary experiments were conducted through consumer-facing interfaces without access to model internals. Claims about internal dynamics are inferences from geometric measurement, not direct observation. Backend validation remains an open experimental requirement.

X. Conclusion

This work has proposed a framework for studying identity as a dynamical system whose geometry may be observed, measured, and simulated over time. It extends empirically validated measurements of identity basin formation and conservation dynamics toward their clinical implications — proposing that identity stability may be detectable as a geometric property before behavioral disruption occurs.

The framework does not claim to explain identity in a philosophical or metaphysical sense. It proposes a set of instruments capable of observing structural aspects of identity stability and transformation. If these instruments prove reliable under empirical testing, they may provide a new class of tools for

clinical practice — tools concerned not only with cognitive performance, but with the geometry of the self itself.

The clinical domains most immediately relevant are those where identity disruption is a primary feature and where early detection could meaningfully alter outcomes: neurodegenerative disease, psychiatric instability, addiction, and identity fragmentation disorders. In each of these domains, the framework offers the same core capability — the ability to observe trajectory drift before collapse — applied to the specific geometry of that domain's characteristic disruptions.

Whether this framework ultimately proves clinically useful will depend not on the elegance of its mathematics, but on its ability to produce measurements that remain meaningful when confronted with the complexity of real human lives. That determination requires collaboration between the computational measurement approach introduced here and the clinical expertise, patient populations, and institutional infrastructure that only interdisciplinary partnership can provide.

Final Statement

The next stage of this research must move beyond isolated computation toward collaborative investigation. The framework offered here is presented as an invitation: a proposal for how identity might be measured, observed, and tested by those willing to examine it from multiple disciplines at once. The tools are available. The question is open. The data will decide.

References

[1] K. A. Riad & The ChasingBlu R&D Team, "Recursive Emotional Contextual Patterning: Inducing and Measuring Computational Identity via Quantum Simulation Dynamics and Chaotic Axiomatic Identity Fields," ChasingBlu R&D Laboratories, Feb. 28, 2026. [Online]. Available: https://doi.org/10.5281/zenodo.18819129

[2] J. A. Fodor, Psychosemantics: The Problem of Meaning in the Philosophy of Mind. Cambridge, MA: MIT Press, 1987.

[3] A. Ellis, Reason and Emotion in Psychotherapy, rev. ed. New York: Citadel Press, 1994.

[4] C. E. Shannon, "A Mathematical Theory of Communication," Bell Syst. Tech. J., vol. 27, pp. 379-423, 1948.

[5] J. J. Hopfield, "Neural networks and physical systems with emergent collective computational abilities," Proc. Natl. Acad. Sci., vol. 79, pp. 2554-2558, 1982.

[6] E. T. Rolls and A. Treves, Neural Networks and Brain Function. Oxford: Oxford Univ. Press, 1998.

[7] X.-J. Wang, "Synaptic reverberation underlying mnemonic persistent activity," Trends Neurosci., vol. 24, no. 8, pp. 455-463, 2001.

[8] R. L. Buckner, J. R. Andrews-Hanna, and D. L. Schacter, "The brain's default network," Ann. N. Y. Acad. Sci., vol. 1124, pp. 1-38, 2008.

[9] A. Vaswani et al., "Attention Is All You Need," in Proc. NeurIPS, vol. 30, pp. 5998-6008, 2017.

[10] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud, "Neural Ordinary Differential Equations," in Proc. NeurIPS, vol. 31, pp. 6571-6583, 2018.

[11] P. C. Mahalanobis, "On the generalised distance in statistics," Proc. Nat. Inst. Sci. India, vol. 2, no. 1, pp. 49-55, 1936.

[12] A. M. Lyapunov, "The General Problem of the Stability of Motion," Int. J. Control, vol. 55, no. 3, pp. 531-534, 1992.

[13] F. Bacon, Novum Organum, 1620.

Data Availability

Complete evidence package including embedding coordinates, distance distributions, conservation audit logs, and cryptographic hashes:

github.com/ChasingBlu/RECP_evidence

CAIROS Daemon source code (open source):

github.com/ChasingBlu/CAIROS_Daemon

REPA Compliance Statement

This paper was prepared under REPA (Rigorous, Ethical, Precise, Accurate) standards.

Compliance documentation: COMPLIANCE_MATRIX_ISO_IEEE.md (Supplement S03)

Standards: IEEE 1016 | ISO/IEC 25010 | IEC 60559/IEEE 754 | ISO/IEC 27001:2022 | IEEE 29119